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Activity Module Documentation

| Property            | Description                                                   |
|---------------------|---------------------------------------------------------------|
| _ActivityService    | Service for handling activity CRUD operations                 |
| _unitOfWork         | Access to repositories like Users, Departments, Subscriptions |
| _mapper             | AutoMapper for mapping between DTOs and ViewModels            |
| _excelReportService | Service to generate Excel reports with localization           |

ActivityController Actions Overview

| Action                                                            | Purpose / Features                                                                                                                                      |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Index                                                             | Display paginated activities, filter by date/keyword, calculate duration and subscription count, supports Ajax partial views                            |
| AddEdit                                                           | Create/Edit activity, file attachment management (PDF ≤ 5MB), dropdown refill on validation fail, temporary file storage, validation for type/size      |
| Details                                                           | PartialView display of activity details, file existence check, dynamic duration, lightweight modal                                                      |
| MembersActivity                                                   | Display members of an activity, pagination, filtering, combined activity/member ViewModel, supports Ajax                                                |
| Delete                                                            | Delete activity by id, async, admin authorization, redirects to list                                                                                    |
| Print & ExportToExcel                                             | Print/export activities, keyword/date filtering, calculate duration/subscription count, bilingual formatting, dynamic columns, improved date formatting |
| createExcelReport_Download                                        | Download filtered activity data as Excel, DTO conversion, localization, timestamp-based file naming, try-catch error handling                           |
| PrintMembersActivity / createExcelReport_Download_MembersActivity | Print/export subscribed members, includes nationality, age, contact, activity context, bilingual support                                                |

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| Validation / Session | Description                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------|
| File Validation      | PDF only, max 5MB, temporary storage, old file cleanup                                          |
| Model Validation     | Required fields: TitleAr/En, dates, location; start < end validation; proper file path handling |
| Language Support     | Bilingual UI, session-based language, Resource1/2                                               |
| Session Management   | Language preference, temp file paths, improved UX                                               |

ActivityService Documentation

| Property    | Description                                                    |
|-------------|----------------------------------------------------------------|
| _unitOfWork | Access to repositories (Activities, Members, Subscriptions)    |
| FileName    | Default folder/file name for saving attachments ("Activities") |

ActivityService Methods

| Method                       | Purpose / Notes                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------|
| GetAllAsync                  | Retrieve all activities asynchronously                                                             |
| GetByIdAsync                 | Retrieve activity by id, returns null if not found                                                 |
| AddAsync                     | Add activity, optional file upload, transactional save, returns new Id                             |
| UpdateAsync                  | Update activity, optional file replacement, preserves old file paths, uses UpdateValues pattern    |
| DeleteAsync                  | Delete activity, cleanup file attachment, transaction commit                                       |
| GetAllMembersOfActivityAsync | Get members subscribed to activity, eager load Nationality, separate member/subscription retrieval |

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Activity Entity

| Property       | Type                  | Notes / Validation     |
|----------------|-----------------------|------------------------|
| Id             | int                   | Primary key, [Key]     |
| TitleAr        | string?               | Max 200, Arabic        |
| TitleEn        | string?               | Max 200, English       |
| StartDate      | DateOnly?             | Start date of activity |
| EndDate        | DateOnly?             | End date of activity   |
| MinimumAge     | int?                  | Minimum allowed age    |
| Location       | string?               | Max 200                |
| Description    | string?               | Optional description   |
| Achievement    | string?               | Optional achievements  |
| AttachmentPath | string?               | Max 300, file path     |
| Questions      | ICollection<Question> | Initialized empty list |

ActivityVM ViewModel

| Property    | Description         | Validation / Notes        |
|-------------|---------------------|---------------------------|
| Id          | Activity identifier | Auto-managed              |
| TitleAr     | Arabic title        | Required, Max 200         |
| TitleEn     | English title       | Required, Max 200         |
| StartDate   | Activity start date | Required, Not in past     |
| EndDate     | Activity end date   | Required, after StartDate |
| MinimumAge  | Minimum age         | Required                  |
| Location    | Activity location   | Required, Max 200         |
| Description | Details             | Optional                  |
| Achievement | Achievements        | Optional                  |

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| Property       | Description       | Notes          |
|----------------|-------------------|----------------|
| AttachmentPath | File storage path | Max length 300 |
| Attachment     | File upload       | IFormFile      |

| Property          | Type | Description                             |
|-------------------|------|-----------------------------------------|
| IsSubscribed      | bool | Indicates if current user is subscribed |
| ActivityDuration  | int? | Duration in days                        |
| SubscriptionCount | int? | Number of participants                  |

| Property  | Description                                        |
|-----------|----------------------------------------------------|
| Questions | Activity-related questions (IEnumerable<Question>) |

| MembersActivityVM | Description       | Notes                                 |
|-------------------|-------------------|---------------------------------------|
| Activity          | ActivityVM        | Activity information                  |
| Members           | MemberVM list     | Related members                       |
| Subscriptions     | Subscription list | Member subscriptions to this activity |

| SubscribedMemberInActivitiesVM | Description               | Notes                     |
|--------------------------------|---------------------------|---------------------------|
| Activity                       | Selected activity details | ActivityVM                |
| Members_Paginated              | Paginated members         | PaginatedList<MemberVM>   |
| Subscriptions                  | Raw subscription records  | IEnumerable<Subscription> |